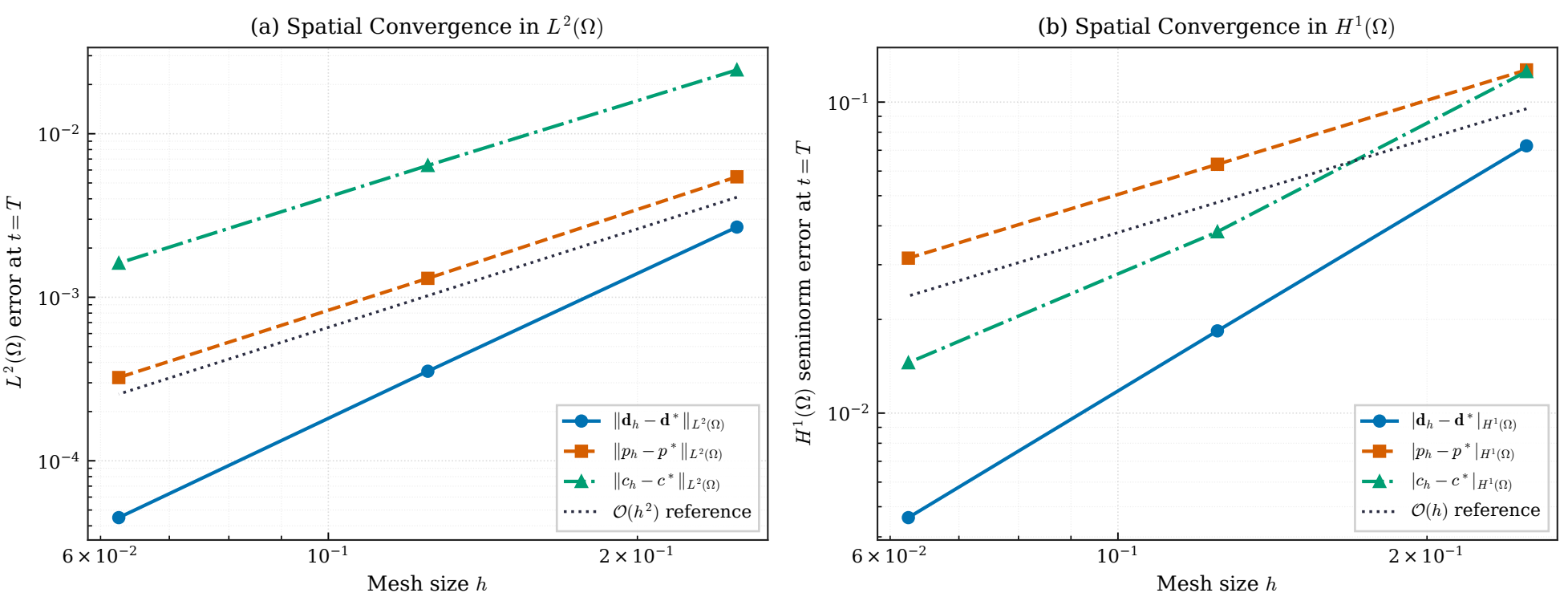
(a) Spatial Convergence in $L^2(\Omega)$
$L^2(\Omega)$ error at $t = T$
10^{-2}
10^{-3}
10^{-4}
6×10^{-2}
10^{-1}
2×10^{-1}
Mesh size h
$\|\mathbf{d}_h - \mathbf{d}^*\|_{L^2(\Omega)}$
$\|p_h - p^*\|_{L^2(\Omega)}$
$\|c_h - c^*\|_{L^2(\Omega)}$
$\mathcal{O}(h^2)$ reference
(b) Spatial Convergence in $H^1(\Omega)$
$H^1(\Omega)$ seminorm error at $t = T$
10^{-1}
10^{-2}
6×10^{-2}
10^{-1}
2×10^{-1}
Mesh size h
$|\mathbf{d}_h - \mathbf{d}^*|_{H^1(\Omega)}$
$|p_h - p^*|_{H^1(\Omega)}$
$|c_h - c^*|_{H^1(\Omega)}$
$\mathcal{O}(h)$ reference